

FIRE PREVENTION AND PROTECTIONFire Prevention

Painting operations of particular concern are those having a fire potential; i.e., paint removal, solvent wipe and paint application by means of spray apparatus. Certain paints, lacquers, varnishes, shellacs, solvents, and thinners are very flammable. These, for the purpose of control, are classified as being flammable. Solvent materials selected to do the residual clean up, after the initial removal, shall have a flash point of 140 degrees F or above.

Spray Painting

Spray painting presents varying degrees of fire hazards, depending on the materials used. Material having a flash point below 140 degrees will be handled very carefully, and precautions are in order even for those having a flash point higher than this.

Sprinklers

Fire suppression sprinklers installed in spray finishing areas shall conform to NFPA 13, provisions for extra hazardous occupancy. Dry chemical, carbon dioxide, or halogenated extinguisher systems may be installed where automatic sprinkler protection is not available.

Extinguishers

SAFETY AND HEALTH PROGRAM

Portable fire extinguishers shall be installed near all paint spraying areas. The Safety and Health Manager shall determine the type of extinguisher that is appropriate.

Ventilation Systems

Ventilation and exhaust systems shall be in accordance with the standard for Blower and Exhaust Systems for Vapor Removal, NFPA 91. Mechanical ventilation shall be in operation while spraying operations are being conducted and for a sufficient time thereafter to assure vapors are completely exhausted. Adequate conditioned make-up air must be provided.

Fan Unit

The fan-rotating element and its casing shall be non-sparking. Ample clearances shall be provided to prevent friction-caused fire hazards. Fan blades shall be mounted on a shaft rigid enough to maintain alignment when the fan is operating under full load.

Exhaust Ducts

Exhaust ducts shall be protected against mechanical damage, properly supported, and will normally have a separation of at least 18 inches from combustible materials. Ducts shall be periodically inspected for accumulation of paint deposits and shall be cleaned as needed.

Exhaust

Air exhaust from spray operations shall be directed so that it will not contaminate make-up air being introduced into the spraying area or other ventilation intakes. Unless the spray booth exhaust duct terminal is from a water-wash spray booth, the terminal discharge point shall be at least 6 feet from any combustible exterior wall or roof.

Motors

Electric motors driving exhaust fans shall not be placed inside booths or ducts. Drive belts shall not enter the duct or booth unless the belt and pulley within the duct or booth are enclosed or guarded.

Storage and Handling

Storage

The quantity of paints, lacquers, thinners, solvents and other flammable and combustible liquids kept near spraying operations shall be the minimum required for operations but shall not exceed 1 day's supply.

Bulk storage of these liquids shall be in a separate building detached from other buildings or in rooms specifically designed and constructed to meet flammable storage room requirements.

No storage of open containers of solvents is permitted. Open containers may only be used for cleaning of painting materials after which the solvent shall be transferred back to a closed container for retention or disposal. Supplies of flammable and combustible liquids shall be stored in approved fire-resistant safety containers equipped with flash screens and self-closing lids. Operations involving water base latex paints are exempt from the above requirements.

Containers

Original closed containers, approved portable tanks, and approved safety cans shall be used for bringing flammable or combustible liquids into spray finishing rooms. Open or glass containers shall not be used.

SAFETY AND HEALTH PROGRAM

Liquid Transfer

The withdrawal of liquids from containers and the filling of containers, including portable mixing tanks, shall be done only in a mixing room or in a spraying area when the ventilating system is in operation. Precautions shall be taken to protect against liquid spillage.

Grounding

Whenever flammable or combustible liquids are transferred from one container to another, both containers shall be effectively bonded and grounded. This practice prevents electrical discharge from the accumulation of static charge because of the transfer process.

Electrical

Electrical Wiring

Electrical wiring and equipment shall conform to the provisions of the National Electrical Code (NFPA 70). Electrical wiring located in spray areas must be rigid metal conduit, Type MI cable, or in metal boxes or fitting containing no taps, splices or terminal connections. There are alternative electrical wiring in options when the location is adjacent to (rather than inside) a spray area (NFPA 33).

Electrical Equipment

Electrical equipment outside of, but within 20 feet horizontally and 10 feet vertically, of any spraying area and not separated from it by partitions extending at least to the boundary of the Division 2 location shall be of non-spark producing design. This equipment shall also conform to the provisions of NFPA 70, for Class I or Class I, Division 2 locations as applicable. If spraying operations are confined to an enclosed spray booth or room, the space adjacent to the booth or room shall be considered non-hazardous except for the space within three feet in all directions from any opening in the booth or room.

Grounding

All metal parts of spray booths and exhaust ducts conveying flammable or combustible liquids or aerated combustible solids shall be electrically grounded.

Accident & Incident Auditing, Investigation, and Reporting

23.0 COMPANY INCIDENT REPORTING SYSTEM

Introduction

The Company Incident Reporting System is designed to:

1. Track and analyze employee injuries and illnesses, property and vehicle damage, as well as serious events or near misses which might have resulted in personal injury, illness, or property and vehicle damage
2. Initiate the worker's compensation process, if necessary
3. Meet regulatory reporting requirements

All incidents (accidents resulting in injury or causing illness to Company employees) and events

(near-miss accidents) shall be reported in order to:

- Establish a written record of factors which cause injuries and illnesses and occurrences (near-misses) which might have resulted in injury or illness but did not, as well as property and vehicle damage.
- Maintain a capability to promptly investigate incidents and events in order to initiate and support corrective and/or preventive action.
- Provide statistical information for use in analyzing all phases of incidents and events involving Company personnel.
- Provide the means for complying with the reporting requirements for occupational injuries and illnesses outlined in Section 19 of the Occupational Safety and Health Act of 1970.

Applicability and Scope

The Incident Reporting System requirements apply to all incidences involving Company personnel arising out of or in the course of employment which results in (or might have resulted in) personal injury, illness, and/or property and vehicle damage.

Incidents (Occupational Injuries and Illnesses)

Injuries and illnesses that require reporting include those injuries and illnesses occurring on the job which result in any of the following: lost work time, restrictions in performing job duties, requirement for first aid or outside medical attention, permanent physical bodily damages, or death. Examples of "reportable injuries and illnesses include, but are not limited to, heat exhaustion from working in hot environments, strained back muscles from moving equipment, acid burns on fingers, etc.

Other incidents requiring reporting include those incidents occurring on the job which result in any of the following: injury or illness, damage to a Government vehicle, fire/explosion, property damage of more than \$100, or chemical releases requiring evacuation of at least that immediate spill area. Examples of reportable incidents include denting the fender of a Government vehicle, spilling 1 liter of sulfuric acid on the floor, and release of pressurized air through a leaking valve into a room.

Examples of "non-reportable" injuries and illnesses include small paper cuts, common colds, and small bruises not resulting in work restrictions or requiring first aid or medical attention. Examples of other non-reportable incidents include, but are not limited to, dropping and breaking an empty glass sample bottle, knocking over 10 milliliters of sulfuric acid in a graduated cylinder under a fume hood and knocking a chair or table over.

Events (Near Misses)